

Context-Aware Autonomous Security Patrol Drone

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Abstract—Autonomous aerial surveillance systems have the potential to improve security monitoring in large and dynamic environments where continuous human supervision is difficult. Traditional surveillance systems rely heavily on static cameras and manually controlled drones, which limits adaptability and intelligent response to suspicious activities. This project presents a context-aware autonomous security patrol drone developed within a Gazebo-based robotics simulation environment. The system integrates reinforcement learning navigation, contextual risk assessment, and autonomous decision-making to enable intelligent patrol behavior.

The proposed framework allows the drone to patrol a pre-defined perimeter, monitor restricted zones, detect suspicious intruder activity, and dynamically adapt its behavior according to environmental context and calculated risk levels. A contextual risk assessment model was developed using multiple behavioral and environmental factors including restricted-zone presence, dwell time, proximity to sensitive areas, and detection confidence. Based on the computed risk score, the drone autonomously switches between patrol, investigation, and alert modes.

For autonomous navigation, a reinforcement learning approach based on Proximal Policy Optimization (PPO) was implemented using Gymnasium and Stable-Baselines3. The agent was trained through reward-based learning to navigate efficiently while avoiding unsafe movement and collisions. Experimental results demonstrate successful perimeter patrol, obstacle-aware navigation, autonomous context switching, and dynamic intruder investigation behavior. The project highlights how reinforcement learning and context-aware decision systems can be integrated within robotics simulation to create intelligent autonomous surveillance systems.

I. INTRODUCTION AND PROBLEM STATEMENT

Autonomous surveillance systems are becoming increasingly important in modern security applications such as campus monitoring, industrial facility protection, warehouse surveillance, and restricted-area security. Traditional surveillance systems primarily rely on static cameras or manually controlled drones that require constant human supervision [10]. While these systems can capture visual information, they are limited in their ability to respond intelligently to changing environmental conditions or suspicious activities.

Static surveillance systems suffer from several major limitations. Fixed cameras provide only limited viewing angles and cannot actively investigate suspicious behavior occurring outside their field of view. Human operators are often required to continuously monitor multiple video feeds, which can lead to delayed responses, reduced attention, and operational inefficiency. Similarly, manually operated drones require continuous remote control and predefined patrol routes, limiting their ability to adapt dynamically to real-time situations.

This project addresses these limitations by developing a context-aware autonomous security patrol drone capable of intelligent decision-making within a simulated robotics environment. The proposed system combines autonomous patrol behavior, contextual risk assessment, and reinforcement learning-based navigation to enable adaptive surveillance operations. Instead of following only fixed scripted behaviors, the drone evaluates environmental and behavioral factors to determine the level of risk associated with detected activity and adjusts its response accordingly.

The developed system operates within a Gazebo simulation environment containing a building, restricted security zone, obstacles, patrol boundaries, and simulated intruders. The drone continuously patrols the environment while monitoring suspicious activity near sensitive areas. A contextual risk assessment mechanism evaluates factors such as dwell time, proximity to restricted zones, and detection confidence to classify situations into low-risk, medium-risk, or high-risk categories. Based on the calculated risk level, the drone dynamically switches between patrol, investigation, and alert behaviors.

To achieve autonomous navigation and adaptive movement, reinforcement learning was implemented using the PPO algorithm through Stable-Baselines3 and Gymnasium. The reinforcement learning agent learns efficient navigation strategies through reward-based training, enabling obstacle-aware movement and intelligent goal-directed navigation. The overall objective of this project is to demonstrate how reinforcement learning, contextual reasoning, and robotics simulation can be integrated to create an intelligent autonomous aerial surveillance framework.

II. IMPORTANCE AND APPLICATIONS

Autonomous surveillance systems are becoming increasingly important in environments that require continuous monitoring, rapid response, and adaptive decision-making. Traditional surveillance systems rely heavily on static cameras and human operators, which can result in limited coverage, delayed response times, and reduced adaptability in dynamic situations. Autonomous aerial drones provide a more flexible and intelligent alternative by enabling continuous patrol, active monitoring, and autonomous investigation of suspicious activity.

The proposed context-aware patrol drone framework has several important real-world applications:

- **Campus Security:** Monitoring university campuses, parking lots, and restricted-access buildings for suspicious activity and unauthorized access.
- **Industrial Monitoring:** Patrolling factories, warehouses, and industrial facilities where continuous surveillance and safety monitoring are required.
- **Restricted-Zone Surveillance:** Protecting sensitive areas such as research facilities, airports, military compounds, and secure infrastructure.
- **Autonomous Perimeter Patrol:** Performing continuous perimeter monitoring without requiring constant human supervision.
- **Smart Surveillance Systems:** Supporting future intelligent security systems capable of adaptive decision-making and context-aware response.

This project demonstrates how reinforcement learning, contextual risk assessment, and robotics simulation can be integrated into a unified intelligent surveillance framework capable of adaptive autonomous behavior.

III. TECHNICAL CHALLENGES

Developing an autonomous context-aware surveillance drone involves several technical challenges related to perception, decision-making, navigation, and environment interaction. Unlike static surveillance systems, an autonomous drone must continuously analyze environmental conditions and dynamically adapt its behavior in real time.

One major challenge is autonomous navigation in dynamic environments. The drone must move safely within the environment while avoiding obstacles such as buildings and environmental objects. Efficient navigation becomes more difficult when the drone must simultaneously patrol the environment and respond to suspicious activity.

Another challenge is contextual risk assessment. The system must determine whether observed activity represents normal behavior or potentially suspicious behavior. This requires combining multiple environmental and behavioral factors such as restricted-zone presence, dwell time, proximity to sensitive areas, and detection confidence into a single interpretable risk score.

Behavior adaptation is also an important challenge. The drone should not rely entirely on fixed scripted movement patterns. Instead, it must dynamically switch between patrol, investigation, and alert behaviors depending on the current situation and calculated risk level.

Integrating reinforcement learning into the surveillance framework introduces additional complexity. The reinforcement learning agent must learn efficient navigation strategies through reward-based training while minimizing unsafe movement and collisions. Designing an effective reward system and stable training environment is critical for obtaining reliable autonomous behavior.

Finally, integrating multiple components such as Gazebo simulation, contextual decision-making, reinforcement learning navigation, and surveillance logic into a unified framework

required careful system design and testing to ensure stable and realistic autonomous operation.

IV. SYSTEM OVERVIEW AND ARCHITECTURE

The proposed system was designed as a context-aware autonomous surveillance framework capable of monitoring a secured environment, detecting suspicious activity, evaluating contextual risk, and adapting drone behavior dynamically. The overall architecture combines robotics simulation, contextual decision-making, and reinforcement learning-based navigation into a unified intelligent surveillance system.

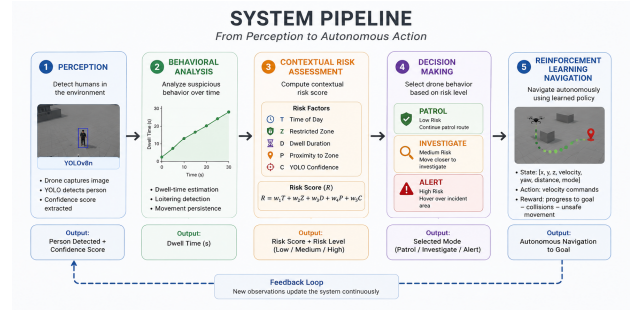


Fig. 1. Overall system pipeline for the context-aware autonomous surveillance framework.

The system pipeline consists of five major stages:

- 1) **Perception**
- 2) **Behavioral Analysis**
- 3) **Contextual Risk Assessment**
- 4) **Decision Making**
- 5) **Reinforcement Learning Navigation**

During operation, the drone continuously patrols the simulated environment while monitoring the intruder's position and activity near restricted areas. The perception stage provides environmental awareness and simulated detection confidence values used for risk estimation. Behavioral analysis evaluates factors such as dwell duration and movement persistence near sensitive areas.

The contextual risk assessment module combines multiple environmental and behavioral factors into a weighted risk score. Based on the calculated score, the system classifies the situation into low-risk, medium-risk, or high-risk categories.

The decision-making module then determines the appropriate drone behavior:

- **Low Risk:** Continue perimeter patrol
- **Medium Risk:** Investigate suspicious activity
- **High Risk:** Move directly toward the high-risk area for closer monitoring

For autonomous movement and navigation, reinforcement learning was implemented using the Proximal Policy Optimization (PPO) algorithm through Gymnasium and Stable-Baselines3. The reinforcement learning agent learns navigation behavior through reward-based optimization, allowing the drone to move efficiently toward target locations while avoiding unsafe movement.

The overall system architecture enables adaptive autonomous surveillance behavior instead of relying entirely on fixed scripted patrol patterns.

V. SIMULATION ENVIRONMENT SETUP

The project was implemented within a Gazebo-based robotics simulation environment designed to represent a secured surveillance area [2]. The environment was created to simulate realistic autonomous patrol scenarios involving restricted zones, obstacle-aware navigation, and intruder monitoring.

The simulated environment contains a central secured building surrounded by a predefined perimeter patrol route. A high-security entrance zone was placed near the building entrance to represent a sensitive restricted area requiring additional monitoring priority. The drone continuously patrols the perimeter while monitoring activity occurring near this protected zone.

Several environmental components were included in the simulation:

- A central secured building
- A high-security entrance zone
- Perimeter patrol boundaries
- Environmental obstacles and trees
- A simulated autonomous drone
- A moving intruder actor

The drone model operates within the Gazebo world and dynamically changes its movement behavior depending on the contextual risk level calculated by the system. The intruder follows predefined movement paths toward the restricted entrance area in order to simulate suspicious activity scenarios.

The environment was developed using Gazebo simulation together with Python-based control modules. The system was implemented and tested using Visual Studio Code as the primary development environment. Reinforcement learning functionality was integrated using Gymnasium and Stable-Baselines3, while numerical computations and environment state handling were performed using NumPy.

The simulation environment provided a safe and flexible platform for testing autonomous patrol behavior, contextual risk assessment, reinforcement learning navigation, and dynamic decision-making without requiring physical drone hardware.

VI. PERCEPTION AND BEHAVIORAL ANALYSIS

The perception and behavioral analysis components were designed to provide environmental awareness and identify potentially suspicious activity within the surveillance environment. These modules allow the drone to monitor intruder behavior and support context-aware decision-making during patrol operations.

The perception stage focuses on detecting the presence of a person within the monitored environment. The system architecture was designed around a lightweight YOLOv8n-based human detection approach suitable for robotics applications and CPU-based systems [5]. Detection confidence values

produced by the perception stage are used as one of the inputs for contextual risk assessment.

In the implemented simulation framework, the intruder's position and activity within the Gazebo environment are continuously monitored to simulate surveillance perception behavior. The system estimates detection confidence values based on the intruder's proximity to sensitive areas and environmental context.

Behavioral analysis was implemented to evaluate suspicious behavior over time rather than relying only on single-frame detection events. The system primarily focuses on dwell-time estimation and loitering behavior near restricted zones.

The behavioral analysis module monitors:

- Intruder presence near sensitive areas
- Dwell duration within monitored regions
- Movement persistence near the restricted zone

Longer dwell durations near the protected entrance area increase the calculated contextual risk score and may trigger investigation or alert behavior from the drone. This allows the system to distinguish between normal movement and potentially suspicious lingering activity.

The dwell-time estimation logic continuously tracks how long an intruder remains near sensitive areas, as shown in Figure 2.

```
# — Dwell-time estimation

detection_counts: dict = defaultdict(int)  #
zone_key -> cumulative frames

def update_dwell(zone_key: str, detected: bool, fps:
int = SIMULATED_FPS) -> float:
    """Increment detection counter for a zone and
    return estimated dwell time.
    """
    if detected:
        detection_counts[zone_key] += 1
    return detection_counts[zone_key] / fps
```

Fig. 2. Implementation of dwell-time estimation logic used for behavioral analysis.

By combining perception-based confidence estimation with behavioral analysis, the surveillance framework is able to support more adaptive and context-aware autonomous monitoring behavior.

VII. CONTEXTUAL RISK ASSESSMENT

One of the core components of the proposed surveillance framework is the contextual risk assessment system. The purpose of this module is to evaluate the severity of observed activity within the environment and determine how the drone should respond to different situations.

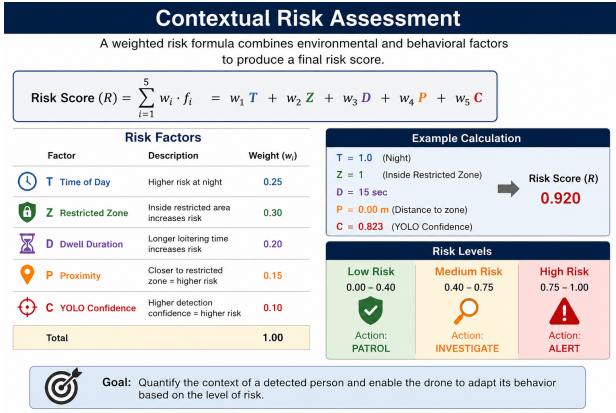


Fig. 3. Contextual risk assessment framework combining environmental and behavioral factors to compute the final risk score.

As shown in Figure 3, the contextual risk assessment module combines multiple weighted environmental and behavioral factors to classify situations into low-risk, medium-risk, and high-risk categories.

Instead of treating all detected activity equally, the system combines multiple environmental and behavioral factors into a single weighted contextual risk score. This enables the drone to adapt its behavior dynamically depending on the current situation and surrounding context.

The contextual risk score is computed using the following weighted formulation:

$$R = w_1 T + w_2 Z + w_3 D + w_4 P + w_5 C$$

where:

- T = time-of-day factor
- Z = restricted-zone presence
- D = dwell duration
- P = distance to the restricted zone
- C = detection confidence

Each factor contributes to the final risk score with different weighting values depending on its importance within the surveillance environment.

The system continuously monitors intruder behavior near the high-security entrance zone. Longer dwell durations, closer proximity to restricted areas, and higher detection confidence values increase the overall risk level.

Based on the computed score, the system classifies the situation into one of three categories:

- **Low Risk** — Continue normal perimeter patrol
- **Medium Risk** — Switch to investigation mode
- **High Risk** — Enter alert mode and move toward the high-risk area

This contextual risk assessment mechanism enables the drone to perform adaptive surveillance behavior instead of relying entirely on fixed patrol logic. The module improves the system's ability to prioritize suspicious activity and respond intelligently to changing environmental conditions.

VIII. DECISION-MAKING SYSTEM

The decision-making module controls how the drone responds to different surveillance situations based on the contextual risk level calculated by the risk assessment system. Instead of relying on fixed scripted patrol behavior, the drone dynamically adapts its actions according to environmental conditions and observed intruder activity.

The system classifies each situation into one of three risk categories: low risk, medium risk, and high risk. Each category corresponds to a different autonomous drone behavior.



Fig. 4. Adaptive drone behavior across low-risk, medium-risk, and high-risk surveillance scenarios.

As shown in Figure 4, the drone dynamically changes its operational mode according to the calculated contextual risk level.

For low-risk situations, the drone continues normal perimeter patrol behavior. During this mode, the drone follows predefined patrol boundaries while continuously monitoring the environment for suspicious activity near the restricted zone.

When the contextual risk score increases to a medium-risk level, the drone switches into investigation mode. In this state, the drone moves closer to the suspicious activity or intruder position in order to perform more focused monitoring and surveillance.

For high-risk situations, the drone enters alert mode and moves directly toward the high-risk location for closer observation. This behavior allows the system to prioritize potentially dangerous situations occurring near the protected entrance zone.

The decision-making framework enables adaptive autonomous surveillance behavior by continuously updating the drone's operational mode according to the calculated contextual risk level. This approach improves system responsiveness and allows the drone to react intelligently to changing environmental conditions instead of relying entirely on static patrol logic.

IX. REINFORCEMENT LEARNING NAVIGATION

Autonomous navigation within the surveillance environment was implemented using reinforcement learning. The objective of the reinforcement learning module was to enable the drone to learn efficient navigation behavior through reward-based optimization rather than relying entirely on manually scripted movement patterns.

The reinforcement learning environment was developed using Gymnasium, while the Proximal Policy Optimization (PPO) algorithm was implemented using Stable-Baselines3 [1]. PPO was selected because of its stability, efficient policy optimization capabilities, and suitability for continuous autonomous navigation tasks within robotics environments [3], [4].

The reinforcement learning problem can be represented as a Markov Decision Process (MDP):

$$M = (S, A, P, R, \gamma)$$

where:

- S represents the environment states
- A represents the available actions
- P represents state transition probabilities
- R represents the reward function
- γ represents the discount factor

The drone receives environmental state information including its position, target location, movement status, and behavior mode. Based on the current state, the PPO agent selects navigation actions that move the drone toward the desired target while avoiding unsafe movement.

The reward system was designed to encourage efficient and safe navigation behavior. Positive rewards were assigned when the drone moved closer to the target location or successfully reached the navigation goal. Negative rewards were assigned for collisions, inefficient movement, and unsafe navigation behavior.

The overall reward formulation can be represented as:

$$r_t = r_{progress} + r_{goal} - r_{collision} - r_{control}$$

where the progress reward is computed as:

$$r_{progress} = \alpha(d_{t-1} - d_t)$$

where d_t represents the current distance to the target location.

During training, the PPO agent continuously interacted with the environment over multiple episodes and gradually improved its navigation policy through reward optimization. Over time, the agent learned autonomous goal-directed navigation behavior capable of supporting patrol, investigation, and alert operations within the surveillance environment.

The reinforcement learning navigation module enabled the drone to perform adaptive autonomous movement while supporting the overall context-aware surveillance framework [6].

X. IMPLEMENTATION DETAILS

The proposed surveillance framework was implemented using a combination of robotics simulation, reinforcement learning, and Python-based autonomous control modules. The overall system was designed and tested within a Gazebo simulation environment to demonstrate adaptive autonomous surveillance behavior.

The simulation environment and surveillance world were developed using Gazebo, which provided the 3D robotics simulation platform for testing patrol behavior, obstacle-aware navigation, and intruder monitoring scenarios. The environment included a secured building, a high-security entrance zone, environmental obstacles, perimeter patrol boundaries, a drone model, and a simulated intruder actor.

The system logic and autonomous control modules were implemented using Python. The project was developed and tested primarily within Visual Studio Code. Numerical computations, environmental state handling, and reward calculations were implemented using NumPy.

The contextual risk assessment module continuously monitored intruder activity and evaluated multiple environmental and behavioral factors including restricted-zone presence, dwell duration, distance to sensitive areas, and detection confidence values. Based on the calculated risk score, the system dynamically classified situations into low-risk, medium-risk, or high-risk categories.

The decision-making module controlled the operational mode of the drone according to the current contextual risk level. Depending on the calculated risk category, the drone autonomously switched between patrol, investigation, and alert behaviors.

Reinforcement learning navigation was implemented using Gymnasium and Stable-Baselines3. A PPO-based reinforcement learning agent was trained to perform autonomous navigation using reward-based optimization. The training process encouraged efficient movement toward target locations while penalizing unsafe navigation and collision behavior.

The Gazebo controller module connected the reinforcement learning navigation system with the surveillance environment by continuously updating drone movement, intruder movement, patrol behavior, and contextual decision-making during simulation execution.

The overall implementation demonstrates the integration of robotics simulation, contextual reasoning, autonomous decision-making, and reinforcement learning navigation into a unified intelligent surveillance framework. The Gazebo simulation environment was created with assistance from ChatGPT [11].

XI. EXPERIMENTAL SETUP AND DEMO SCENARIOS

The final system was implemented and tested in a Gazebo-based simulation environment rather than AirSim due to setup and compatibility limitations on the macOS development system. The implemented system used a pipeline comprised of computer vision, contextual risk scoring, logging, and reinforcement learning navigation. In Gazebo, the drone was

represented by a patrol drone model, which had its position updated by Gazebo service calls.

For the perception component, YOLOv8 was used to determine the presence or absence of a person in a test image. The detector returns whether a person is present along with their confidence. This score is then input into the contextual risk score. The scenario script ran 4 security cases: normal visitor, suspicious activity close to the restricted zone, loitering close to the restricted zone, and an overnight breach. These differed by location of person, time of day, type of zone, amount of dwell time and distance to the restricted zone. The risk system employed a weighted model based on time of day, sensitivity of the zone, dwell time, proximity to the restricted zone and YOLO confidence. Any score under 0.40 was assigned as patrol mode, scores between 0.40 and 0.75 assigned to investigate mode and any score above 0.75 was assigned as alert mode.

For the reinforcement learning portion, the system employed a Gymnasium environment and Stable-Baselines3 PPO algorithm. The state was comprised of the drone's position, velocity, yaw and distance to goal along with its current mode. The actions controlled the x, y and z velocity as well as the yaw rate. The reward function trained the drone to travel to the target while also punishing boundary collisions and large action commands. The PPO system was trained for 512 steps and the final model was saved as `ppodronepatrol`.

The following primary demo scenarios were implemented:

A. Normal Patrol Scenario

In the normal patrol scenario, no immediate threat was present near the restricted zone. The drone continuously performed perimeter patrol behavior while monitoring the environment for suspicious activity. During this scenario, the contextual risk level remained low and the drone stayed in patrol mode.

B. Suspicious Activity Investigation Scenario

In this scenario, the intruder approached the high-security entrance zone and remained near the protected area for an extended duration. The contextual risk score increased due to restricted-zone presence and dwell-time estimation. As a result, the drone switched from patrol mode into investigation mode and moved closer to the suspicious activity for additional monitoring.

C. High-Risk Alert Scenario

In the high-risk scenario, the intruder remained near the restricted entrance zone long enough to trigger a high contextual risk score. The drone entered alert mode and moved directly toward the high-risk area for closer surveillance and monitoring.

These experimental scenarios demonstrated the ability of the system to perform adaptive autonomous surveillance behavior using contextual risk assessment and reinforcement learning-based navigation.

XII. RESULTS AND EVALUATION

Figure 5 demonstrates the final surveillance framework operating within the Gazebo environment. The drone dynamically switches between investigation and alert behaviors according to the calculated contextual risk score while monitoring intruder activity near the restricted zone.

The YOLO detection component was able to detect the person present in the test image with a confidence of 0.823, indicating it could adequately feed the risk assessment component with useful information.

The scenario testing produced the expected outcomes in all 4 cases; in the normal visitor scenario, a person was in a public area, during the day, 17.63 meters from the restricted area and was there for 5 seconds, resulting in a risk score of 0.258 and the drone remained in patrol mode. When the person was 2 meters from the restricted area for 10 seconds (suspicious activity), the risk score was 0.470, and the drone shifted to investigate mode. For the last two scenarios (loitering, breach), a person was inside the restricted area at night. These scenarios generated risk scores of 0.920 (loitering) and 0.982 (breach), placing them both in alert mode, which kept the drone in a hovering position over the anomaly.

The PPO-based navigation component demonstrated goal-directed autonomous movement within the simulated environment. During testing, the drone moved toward target locations while reducing its distance to the assigned goal. As the drone approached the target region, the system dynamically transitioned between patrol and investigation behaviors according to the contextual risk level. The reward function produced positive rewards for movement toward the target and negative rewards for unsafe movement or boundary violations. Overall, the system successfully integrated person detection, contextual risk assessment, adaptive drone behavior, logging, and PPO-based navigation within a unified surveillance framework [7]–[9].

XIII. WORK DISTRIBUTION

The project was completed collaboratively by both team members, with both contributors actively involved in Gazebo environment setup, system development, testing, debugging, experimentation, and presentation preparation throughout the project life cycle.

Rucha Abhay Panhale primarily contributed to the reinforcement learning navigation framework, PPO agent integration, contextual risk assessment system, autonomous decision-making logic, Gazebo surveillance environment development, patrol and investigation behavior implementation, system integration, and report preparation. Additional contributions included development of the high-security monitoring logic, reinforcement learning environment setup, and evaluation of autonomous surveillance behavior within the simulation environment.

Brianna Jules primarily contributed to the overall surveillance system design, perception and behavioral analysis framework, reinforcement learning navigation framework, simulation experimentation, system architecture development, en-

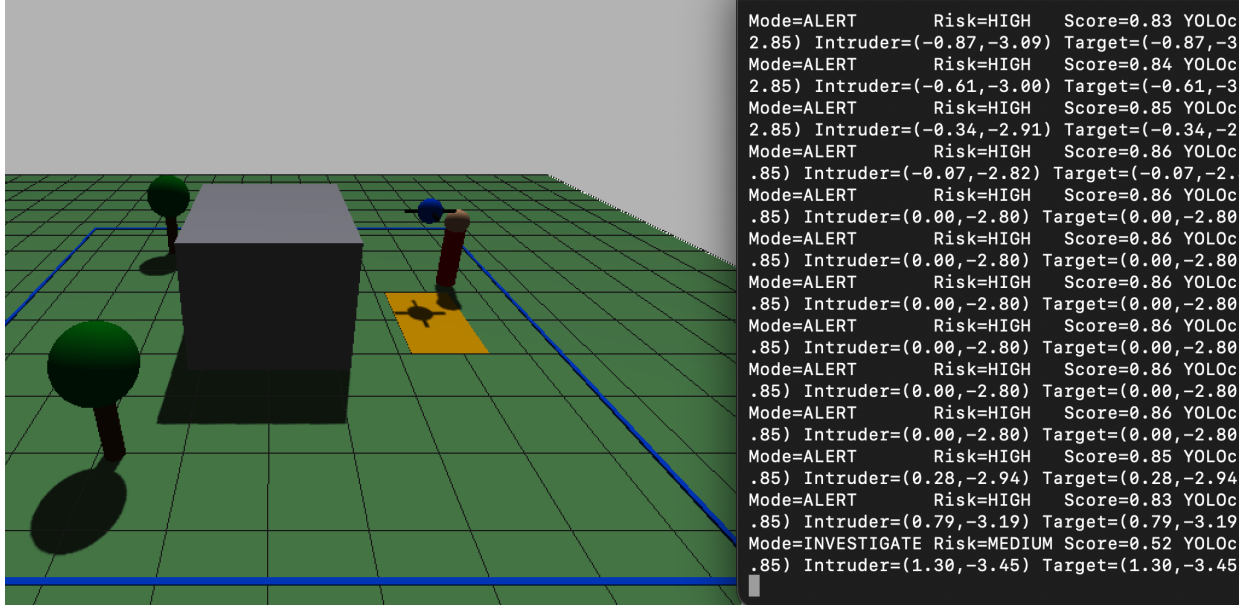


Fig. 5. Experimental results showing autonomous drone surveillance behavior, contextual risk assessment, and adaptive mode switching within the Gazebo simulation environment.

vironmental scenario design, and presentation preparation. Additional contributions included refinement of the contextual surveillance pipeline, evaluation of system behavior across different risk scenarios, and testing of autonomous drone responses within the Gazebo environment.

Both team members collaboratively worked on project planning, implementation, debugging, simulation setup, testing, presentation development, and refinement of the final context-aware autonomous surveillance framework.

XIV. LIMITATIONS

Although the proposed surveillance framework successfully demonstrated context-aware autonomous behavior within the Gazebo simulation environment, several limitations remain in the current implementation.

The system was tested only in simulation and was not deployed on physical drone hardware. Real-world deployment would introduce additional challenges such as sensor noise, communication delays, hardware limitations, and environmental uncertainty.

The current perception framework uses simulated detection confidence values rather than a fully integrated real-time computer vision pipeline. In addition, the reinforcement learning navigation system uses a simplified navigation abstraction instead of full low-level aerial drone dynamics.

The system currently supports only a single drone and a single intruder within the environment. More complex surveillance scenarios may require multi-drone coordination and larger-scale environments.

Despite these limitations, the project successfully demonstrates the integration of reinforcement learning, contextual risk assessment, and robotics simulation within an intelligent autonomous surveillance framework.

XV. FUTURE WORK

Several improvements can be made to extend the capabilities of the proposed surveillance framework. Future work may include integrating a real-time YOLO-based computer vision pipeline for live intruder detection and tracking within the simulation environment.

The system can also be expanded to support multi-drone coordination for large-scale autonomous surveillance and perimeter monitoring tasks. More advanced deep reinforcement learning algorithms and realistic aerial drone dynamics could further improve navigation performance and autonomous decision-making.

Additional future enhancements may include SLAM-based autonomous mapping, real-time camera processing, larger dynamic environments, and deployment on physical drone hardware for real-world surveillance applications.

XVI. CONCLUSION

This project presented a context-aware autonomous security patrol drone framework developed within a Gazebo-based robotics simulation environment. The proposed system combined contextual risk assessment, autonomous decision-making, and reinforcement learning navigation to enable adaptive surveillance behavior.

The surveillance framework successfully demonstrated autonomous perimeter patrol, dynamic investigation behavior, contextual risk assessment, and reinforcement learning-based navigation within multiple simulated security scenarios. The drone was able to dynamically switch between patrol, investigation, and alert modes according to the calculated contextual risk level.

Overall, the project demonstrates the feasibility of integrating robotics simulation, contextual reasoning, and rein-

forcement learning into an intelligent autonomous surveillance system capable of adaptive decision-making and autonomous monitoring behavior.

REFERENCES

- [1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [2] N. Koenig and A. Howard, "Design and Use Paradigms for Gazebo, An Open-Source Multi-Robot Simulator," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2004.
- [3] M. Towers et al., "Gymnasium," GitHub Repository, 2023. [Online]. Available: <https://github.com/Farama-Foundation/Gymnasium>
- [4] A. Raffin et al., "Stable-Baselines3: Reliable Reinforcement Learning Implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.
- [5] G. Jocher et al., "YOLO by Ultralytics," GitHub Repository, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [6] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press, 2018.
- [7] Y. Sheng et al., "UAV Autonomous Navigation Based on Deep Reinforcement Learning," *Drones Journal*, 2024.
- [8] T. Hickling et al., "Deep Reinforcement Learning Based Autonomous Decision-Making for Cooperative UAVs," *arXiv preprint*, 2025.
- [9] N. Khial et al., "DRONE-RL: Dynamic Reinforcement Learning for Online UAV Navigation," *Knowledge-Based Systems*, 2025.
- [10] A. Chriki, H. Touati, H. Snoussi, and F. Kamoun, "UAV-Based Surveillance System: An Anomaly Detection Approach," *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2020.
- [11] OpenAI, "ChatGPT," 2026. [Online]. Available: <https://chat.openai.com/>